TECHNOLOGY STACK

Comic Craft – AI Comic Story Creator using Gemini Models

Date	15 March 2026
Team ID	SWTID-2026-8237
Project Name	Comic Craft – AI Comic Story Creator using Gemini Models

Maximum Marks: 2 Marks

Technology Stack: The following technologies are selected based on the Comic Craft project documentation. The stack supports the web interface, Fast API backend, AI-based story generation, image generation, PDF export, local deployment, and development workflow.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML, CSS, Jinja2	Used to build the interactive Comic Craft web interface. HTML/CSS provide the page structure and styling, while Jinja2 dynamically renders the comic input, preview, and export-success pages. The project uses index.html, comic_preview.html, and export_success.html.
2	**Backend / Server-Side**	Python, unicorn	Python provides the application language; Fast API handles server-side operations, routing, form/JSON input processing, AI workflow coordination, and API endpoints. Unicorn runs the Fast API application locally with the documented --reload development workflow.
3	**Database / Data Storage**	Not specified / No database documented	The supplied Comic Craft project documentation does not specify a database such as MySQL, MongoDB, or PostgreSQL. The documented version focuses on single comic generation without user account management. Generated panel images are stored under static/panels and exported PDFs under static/exports.
4	**Cloud / Hosting / Deployment**	Local deployment with Unicorn; cloud hosting not specified	The documented deployment is local using a Python virtual environment and Unicorn. The project states that cloud deployment may be considered in the future, but no specific cloud provider is selected in the supplied documentation.

5	**Version Control & CI/CD**	Git	Git is listed in the project prerequisites for version control. No specific CI/CD platform or automated pipeline is documented in the supplied project material.
6	**Third-Party APIs / Other Tools**	Google Gemini API, Hugging Face Diffusers, Stable Diffusion, Py Torch, FPDF, Pillow	Gemini Flash generates structured 5-panel comic outlines; Gemini Pro generates detailed narration and character dialogue; Stable Diffusion through Hugging Face Diffusers generates comic-style illustrations; Py Torch supports the image-generation stack; FPDF creates the downloadable multi-page comic PDF; Pillow is included among the required libraries.

Comic Craft Technology Flow

Layer / Function	Technology / Component
User Interface	HTML + CSS + Jinja2
Web/Application Server	Fast API + Unicorn
Story Outline Generation	Google Gemini 1.5 Flash
Story Narration & Dialogue	Google Gemini 1.5 Pro
Comic Image Generation	Stable Diffusion via Hugging Face Diffusers
Layout Construction	layout_builder.py
PDF Export	FPDF via exporters.py
Generated Assets	static/panels and static/exports
Development Environment	Python + pip + virtual environment + Git

Source note: This document is prepared from the supplied Comic Craft project document and the supplied Technology Stack template. Where the project source does not specify a technology, the field is marked as not specified rather than introducing an unsupported technology.